

Requirements

- Functional
 - Users can sign up, log in, and keep separate profiles under one account.
 - Users can browse content on a dashboard and search by title, genre or cast.
 - Users can stream content, with pause, resume and seek.
 - Video quality adjusts automatically to the available network speed.
 - Users can view watch history and resume a title on any device.
 - Users can save titles to a personal list.
 - Users can subscribe to a plan and manage payment.
- Non-Functional
 - **Availability:** 99.99%+, no user visible downtime on playback.
 - **Scalability:** Millions of concurrent streams, with room to grow.
 - **Low latency:** API under 300 ms, playback starting in 2 to 3 seconds.
 - **Fault tolerance:** No single point of failure in the request path.
 - **Durability:** Watch progress and account data are never lost.
 - **Global reach:** Multi region deployment with automatic failover.
 - **Security:** Encrypted transport, authentication, authorization, DRM on video.

Note: billing needs strong consistency, but watch progress can be eventually consistent. This split is why different parts of the system use different databases.